Test Physics Paper

Time allowed: 20 minutes

Total Marks	/ 10

INSTRUCTIONS

Answer all the questions in the spaces provided.

1 **(a)** State Newton's second law of motion.

[2]

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(b) Calculate the resultant force on a 2.0 kg mass accelerating at 3.0 m/s.

[2]

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2 Draw the path of the ray through the glass block on Figure 1.

[3]

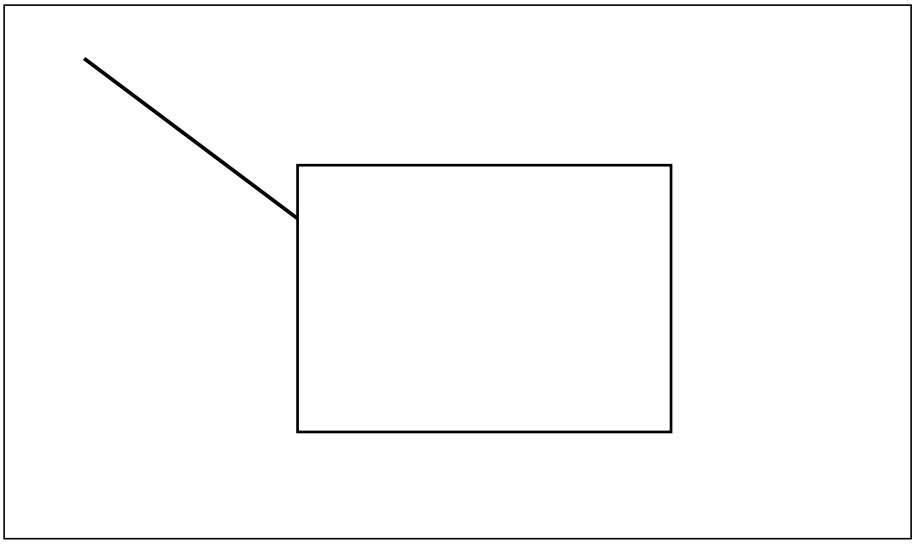

Figure 1

3 Which of these is a vector quantity?

A. mass

B. speed

C. velocity

D. energy

[1]

4 Explain why a skydiver reaches terminal velocity.

[2]

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5 Complete the truth table for an AND gate.

[2]

A	B	Q
0	0	
0	1	
1	0	
1	1	

6 Draw a logic circuit for Q = NOT (A AND B) using one logic gate.

[2]

7 **(a)** State the purpose of each of these three buses.

Data bus .....

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Address bus .....

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Control bus .....

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[3]

(b) Describe how a multi-level feedback queue works.

[3]

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(c) Explain why hardware devices require a device driver.

[2]

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